

Palacian Tudor-Bogdan

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EDUCATION

Technical University of Cluj-Napoca

Bachelor's degree in Computer Science

- Currently in the 3rd year of study

Cluj-Napoca, RO

Oct. 2022 – July 2026

PROJECTS

Academic Management System — *Flask, PostgreSQL, SQLAlchemy, Bootstrap, JavaScript*

- Developed a web-based education platform (UniTasker) with role-based access control for students, professors, and administrators.
- Implemented course management, assignment submission, grading, and academic scheduling with conflict detection.
- Built with Flask backend, PostgreSQL database, SQLAlchemy ORM, and responsive Bootstrap frontend.

PC Benchmarking Suite — *C, Python, Real-time System Monitoring, Hardware Performance Analysis*

- Developed a comprehensive PC benchmarking application with a Python GUI frontend and C backend to evaluate critical hardware performance metrics.
- Implemented low-level C benchmarks measuring arithmetic computation, logical operations, floating-point operations, and memory access speeds.
- Created real-time system monitoring through direct hardware communication using WMI interface to extract CPU metrics.
- Designed visualization components with dynamic color-coding and historical performance data tracking.

Sensor Anomaly Detection System — *Python, VHDL, AXI4, CUSUM Algorithm, FPGA*

- Developed a hybrid software-hardware anomaly detection system for temperature sensor readings using CUSUM algorithm.
- Created Python data processing pipeline for sensor data analysis and designed modular VHDL hardware implementation.
- Designed a complete hardware implementation using VHDL with modular architecture and industry standard AXI4 streaming interface.

Digital Locker Security System — *VHDL, FPGA, Digital Circuit Design, State Machines*

- Designed and implemented a secure digital locker system with a configurable 3-digit hexadecimal combination lock using VHDL.
- Utilized hierarchical design methodology with modular components including debouncer circuits and SSD controllers.
- Implemented advanced hardware features such as input debouncing, finite state machines, and visual feedback.
- Developed a custom 7-segment display controller for user interface with dynamic visual feedback.

TECHNICAL SKILLS AND CERTIFICATIONS

Programming Languages: C/C++, Python, Java, SQL, VHDL

Frameworks: Flask, Bootstrap, SQLAlchemy

Tools: FPGA, Git, GitLab, Maven

Other Skills: Hardware Design, Embedded Systems, Real-time System Monitoring, State Machines, AXI4, Digital Circuit Design, Databases (PostgreSQL), Object-Oriented Programming, Software Testing

Certifications: Cambridge English C2 Certification

Soft Skills: Problem-solving, Critical Thinking, Analytical Thinking, Communication, Teamwork, Adaptability, Time Management, Perseverance, Attention to Detail, Resilience

EXTRACURRICULAR ACTIVITIES

Music Production & DJing: Active music producer and DJ.

Coding Contests: Participated in various coding competitions during high school and university.

Basketball: Played basketball as an extracurricular activity.